ABDULRAHMAN ALSAADI

MECHATRONICS ENGINEER | MECHANICAL & PRODUCT DESIGN

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PROFESSIONAL SUMMARY

CSWP-certified Mechatronics Engineering graduate with hands-on experience in mechanical design, prototyping, and electromechanical product development. Independently designed, manufactured, and validated a 3-DOF SCARA robotic arm with a verified 492.5 mm reach and 100 g payload, reducing one link's mass by approximately 16.5% through topology optimization. Experienced in DFM, dimensional tolerancing, manufacturing drawings, and iterative prototype testing.

TECHNICAL SKILLS

Mechanical Design & CAD: SolidWorks, Mechanism Design, Dimensional Tolerancing, Fits & Clearances, Bearing Fits
Analysis & Optimization: SolidWorks Simulation (FEA), Topology Optimization, Torque Calculations
Manufacturing & Prototyping: DFM, FDM 3D Printing, Prototyping, Manufacturing Drawings, BOM, Assembly & Fit Validation
Electromechanical Systems: Belt Transmissions, Stepper/Servo/BLDC Motors, ESP32, Arduino, Raspberry Pi 5

SELECTED MECHANICAL DESIGN PROJECTS

WD DRONE TEAM - TEKNOFEST 2026 UAV Competition View Project
Karabük, Türkiye
UAV & Robotics Engineer **Feb 2026 - Jul 2026**

- Contributed to UAV mechanical development through component placement, fastener selection, BOM preparation, prototype assembly, and redesigns to resolve fit and integration issues.
- Performed preliminary structural FEA in SolidWorks on 5 mm carbon-fiber arm plates and 3 mm central-body plates for an 8 kg static-load case using supplier material data to support design assessment.
- Supported UAV assembly and physical testing, including an 8 kg structural load test and five flight-test sessions, with a 7-minute hover carrying a 1 kg payload.

SCARA ROBOTIC ARM - GRADUATION PROJECT View Project
Karabük, Türkiye
Mechatronics & Robotics Engineer **Apr 2026 - Jun 2026**

- Independently designed, manufactured, assembled, and physically validated a 3-DOF SCARA robotic arm, achieving a verified 492.5 mm reach and 100 g payload at full extension.
- Designed custom links, bearing housings, motor interfaces, and a two-stage GT2 belt transmission; integrated NEMA 17 motors and calculated joint torque requirements.
- Reduced Arm Link 1 mass by approximately 16.5% through SolidWorks Simulation and topology optimization while retaining critical mounting and load-bearing regions.
- Produced manufacturing drawings and a BOM; applied DFM, dimensional tolerancing, and bearing-fit design, resolving prototype assembly issues and manufacturing functional PLA+ components using FDM printing and secondary machining operations.

EXPERIENCE

SMART METHODS View Work
Jeddah, Saudi Arabia
Robotics Engineering Intern **Jul 2025 - Aug 2025**

- Designed a 5-DOF articulated robotic-arm concept in SolidWorks, including links, joint structures, motor mounts, and overall assembly layout.
- Redesigned a robotic-arm joint bracket with an integrated bearing housing and evaluated its structural behavior under static loading using SolidWorks Simulation.
- Prepared CAD models for FDM 3D printing and worked with print settings, including infill, supports, and orientation, to produce prototype components.
- Assembled a separate 3D-printed 4-DOF robotic arm with servo actuators and ESP32/Arduino electronics, preparing the prototype for subsequent programming and testing.

WASH THRU View Work
Jeddah, Saudi Arabia
Industrial Automation Panel Assembly Intern **Jun 2025 - Jul 2025**

- Assembled customized industrial control panels from electrical schematics, working with a Siemens PLC, I/O expansion modules, VFDs, contactors, relays, motor-protection devices, energy meters, and terminal components.
- Performed wire preparation, crimping, termination, component placement, and cable management while documenting assembly activities for technical reporting.

EDUCATION

Bachelor of Science (B.Sc.) in Mechatronics Engineering **Jun 2021 - Jun 2026**
Karabük University, Türkiye

CERTIFICATIONS & PROFESSIONAL MEMBERSHIPS

- SOLIDWORKS: Simulation for Finite Element Analysis Course, LinkedIn Learning | Oct 2026
- Engineering Drawings for Manufacturing Course, LinkedIn Learning | Oct 2026
- Engineer Membership (Active), Saudi Council of Engineers (SCE) | Sep 2026
- Certified SOLIDWORKS Professional - Mechanical Design (**CSWP**), Dassault Systèmes | Aug 2026
- Certified SOLIDWORKS Associate - Mechanical Design (**CSWA**), Dassault Systèmes | May 2026

LANGUAGES

Arabic: Native | English: Upper-intermediate | Turkish: Intermediate